Route packing mathematical overview

Adult theorem map | Draft and unpiloted

Setting

A finite directed graph has one Start s and a different Finish t. Every arrow has capacity one. Routes are simple directed s-to-t paths; different routes may share dots but not arrows. A closure removes an arrow, not a dot.

Main theorem

The greatest number of edge-disjoint routes equals the fewest arrows whose removal blocks every route (the unit-capacity max-flow/min-cut theorem). Equivalently, minimize the number of arrows leaving a set S containing s but not t. Incoming arrows do not count. A route may cross a cut more than once.

Certificates and repair

A collection of k routes and a blocking set of k arrows certify both optima. Being unable to add an unused forward route does not certify optimality. A simple residual change-walk uses unused arrows forward and reserved arrows backward; a backward step cancels a reservation. An s-to-t change-walk increases the route count by one after any cycles are removed. If none exists, the reachable dots determine a matching cut.

Across the bands

K-1 constructs small route collections, finds matching closures and investigates a game. Grades 2-3 organizes all small closures and encounters internal bottlenecks and rerouting. Grades 4-5 distinguishes outgoing from incoming cut arrows, cancels several reservations and can justify the general theorem.

Discovery and limits

First try routes and closures; then conjecture a maximum; finally give both sides of a certificate. The guide keeps the longer conservation and decomposition proof after the keyed problems. The theorem here is for one common endpoint pair and edge capacity, not several prescribed pairs or vertex-disjoint routes. Keep the theorem map with adults until children have explored.

Route packing facilitator guide

Library slot 13. Adult guide for final F13-K-v3 (6 pages, Problems 1-6), F13-23-v3 (7 pages, Problems 1-6) and F13-45-v3 (7 pages, Problems 1-7). Draft and unpiloted; the slot is not a scheduled meeting. Middle page 6 and upper page 2 are clean recording copies labelled as extra workspace, not extra numbered problems.

Purpose and entry points

Build a large collection of simultaneous arrow-following routes, then prove it is largest with a small set of closures. A route collection plus an equally sized blocking set is a complete optimality certificate. A locally stuck collection may need parts undone; final routes still follow arrows.

Entry	Prerequisites and reasonable endpoint
K-1	Follow arrows and spoken rules; distinguish edges from dots; count to three. Aim for a matching route/closure pair or a two-player strategy.
Grades 2-3	Read short vertex labels; organize two-element sets; trace paths. Aim for a complete eight-closure list, an interior bottleneck or the revised rerouting challenge.
Grades 4-5	Track used/unused arrows and direction; reason about all routes. Aim for a two-cancellation repair and a cut certificate. The general proof is optional adult-supported work.

Supplies and preparation

For ten children, prepare selected boards printed single-sided on US Letter at 100%, pencils/erasers, three distinct colored pencils per child or pair, and 120 small paper closure markers (12 per child, 15-20 mm). These are proposed supplies, not verified stock. Test a marker at the middle of a short printed edge so it hides neither an arrowhead nor a dot. Pencil slashes can replace markers. Use line patterns or route numbers as well as color. Bring spare paper for a fresh reservation.

Where a route must be undone (K-1 P1 upper; middle P5; upper P1 and P5), colored pencil does not erase and nothing on paper stops two routes sharing an arrow. Draw routes there in pencil with different line patterns, or in wipe-off marker on a sheet protector. This is a prediction; it is untested.

Keep three stable adult-led tables, with bands chosen by readiness. K-1 vertices are unlettered; the reference diagrams in this guide add letters at exactly the same positions for adults. Start is s and Finish is t in written route lists. An edge such as AC means A to C. Trace one example and try the revised two-cancellation case before teaching.

Rules to protect: a route never revisits a dot; two routes may share a dot but never an arrow. Closures remove arrows, not dots. Every arrow has capacity one for the whole collection. Sending pieces at different times is a different problem. A geometric crossing without a dot is not a junction.

Flexible one-hour menu

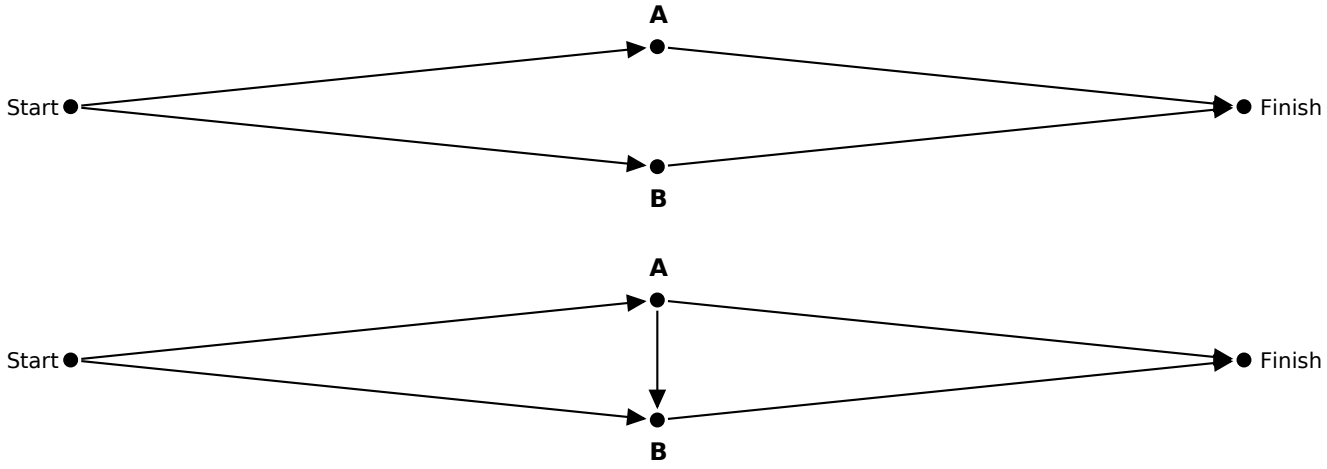
Minutes	Use
0-5	Run around; the children arrive straight from school.
5-9	Shared launch below, then let children handle the closure markers.
9-28	K-1 P1-P3; middle P1-P2; upper P1 or P2.
28-50	Choose one depth: K-1 P4-P5 or game P6; middle P3, P4 or P5; upper P3-P6. Upper P7 is surplus proof work.
50-60	Share a route collection and a closure certificate, or a repair that changed a stuck choice.

Shared launch

Draw a three-arrow chain Start-A-B-Finish. Ask one child to trace a route and another to put a marker on one arrow. Show that closing the arrow prevents every route on this chain. Then draw two branches meeting at a dot and continuing separately. Trace two routes that share the dot, and emphasize that only sharing an arrow is forbidden. Then start a second route along an arrow the first already uses, and ask a partner whether that is allowed: it is not. Keep this under four minutes; leave the maximum search to the tables.

Two reference boards

The diamond has two independent branches. The greedy board adds the vertical A-to-B arrow. K-1 P1 upper, P4 lower and P6 lower use the greedy board; K-1 P2 upper and P6 upper use the diamond.



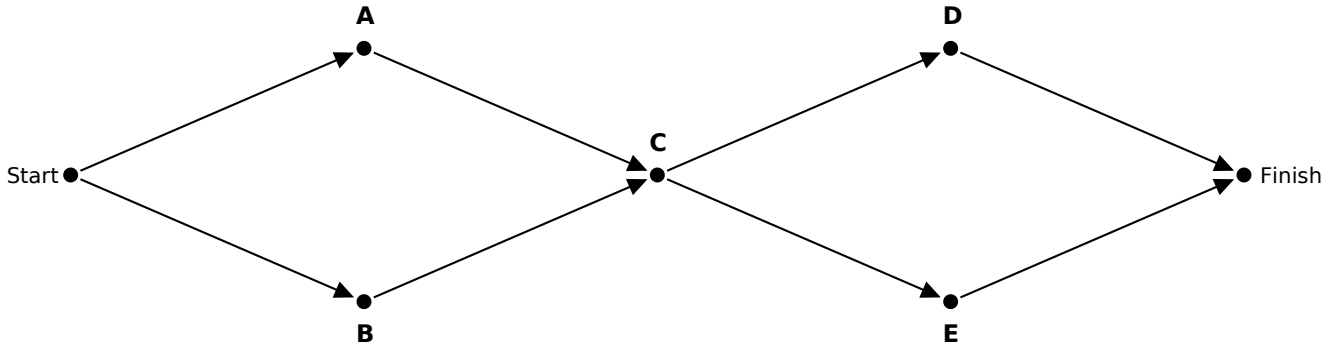
How to use the hints

Let children show what they tried first. Ask the next small question only when they are stuck. For a proposed certificate, ask a partner to try to sneak a route through the closures. Do not replace the search with “close these two arrows.” A matching certificate matters more than finishing pages.

K-1 Problems 1 and 2

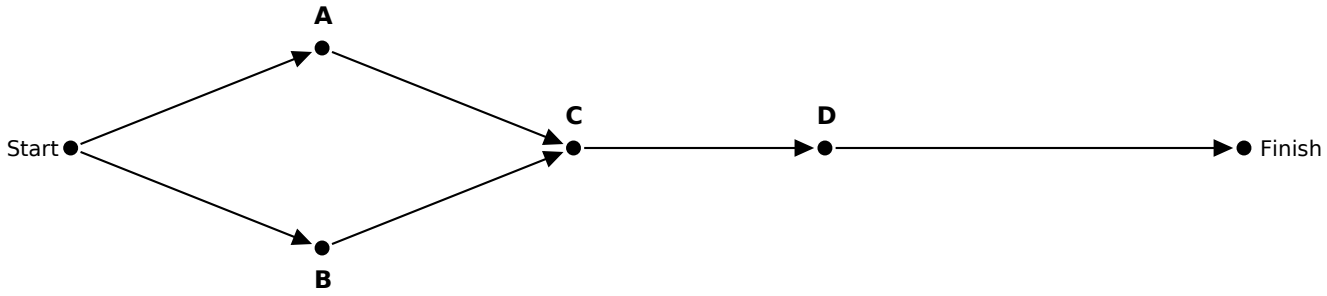
Problem 1 upper and lower

Upper greedy board: maximum 2, uniquely the collection s-A-t and s-B-t. The route s-A-B-t alone is stuck but not largest. Lower bowtie: maximum 2, with exactly two collections: {s-A-C-D-t, s-B-C-E-t} or {s-A-C-E-t, s-B-C-D-t}. Both may meet at C. Closing sA and sB blocks all travel and certifies 2. Hint: can each arrow out of Start serve a different route? If a chosen route blocks another, try changing the first.



Problem 2 upper and lower

Upper diamond: minimum 2. All four minimum closures take one arrow from {sA,At} and one from {sB,Bt}: {sA,sB}, {sA,Bt}, {At,sB}, {At,Bt}. Each branch must be broken. Lower funnel: minimum 1; either CD or Dt stops both routes. There are no other one-arrow solutions because closing an earlier edge leaves the other branch. Hint: trace both routes and look for an arrow they must both use.



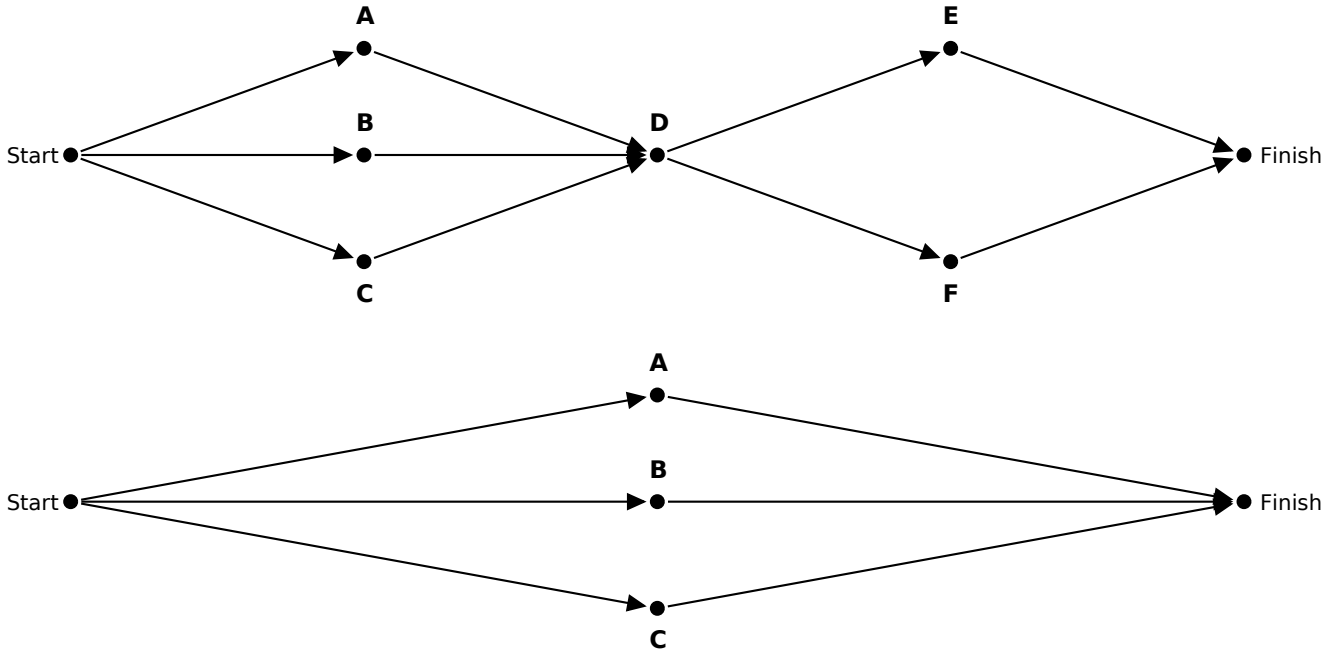
The certificate argument for any band

Suppose k edge-disjoint routes are visible. A set of fewer than k closed arrows cannot hit every route, because a closed arrow belongs to at most one of these routes. Thus at least k closures are necessary. Conversely, if k arrows block every route, every route in any edge-disjoint collection must consume a different one of those arrows, so at most k routes fit. Finding both objects proves both optima without a list of all routes.

K-1 Problems 3 to 6

Problem 3

Upper board below: 2 routes, for example s-A-D-E-t and s-B-D-F-t; close DE and DF. Lower three-branch board: 3 routes, s-A-t, s-B-t and s-C-t; close sA, sB and sC. Every route uses one selected outgoing Start arrow. The page asks for as few markers as possible; ask children to compare that number with the number of routes. Hint: a shared dot may have several incoming arrows but fewer outgoing arrows.



Problem 4

On the upper board, exactly six single closures preserve two routes: sA, sB, sC, AD, BD, CD. Each removes one of three approaches to D, leaving two intact. Any of DE, DF, Et or Ft removes one of only two exits and leaves at most one route. On the lower greedy board, only AB may be removed. Any other closure breaks one of the two disjoint branch routes. Reopen the arrow before each new trial. Hint: sort arrows into “before the meeting dot” and “after it.”

Problem 5

Exactly eight pairs; use the list and completeness proof on page 6. K-1 page 5 has a large working board and ten small recording copies, so the copies do not give the count away. Mark a different successful pair on each; a marker rearrangement alone does not preserve the record. Hint: check that two marked arrows belong to the same pair before comparing records.

Problem 6

Upper diamond: choose second. Any first move breaks one branch; break the other to win immediately. Lower greedy board: choose first and close AB. This leaves the diamond for the opponent, so break the remaining branch after their move. This is the only winning opening: any other first closure leaves all surviving routes sharing one arrow, which the opponent closes at once. Hint: play each opening with roles reversed; ask who can make the last necessary closure, not who closes the most arrows.

Grades 2-3 Problems 1 to 4

Problem 1

Both boards have 2 routes and require 2 closures. Upper bowtie: take s-A-C-D-t and s-B-C-E-t; close {sA,sB}. Lower greedy: take s-A-t and s-B-t; close {sA,sB}. Page 4 explains the certificate; shared C does not consume capacity. Hint: distinguish a dot that is shared from an arrow that is shared.

Problem 2

Maximum and minimum are both 2. Use s-A-D-E-G-H-t and s-C-D-F-G-J-t; close DE and DF (or EG and FG). Start has three outgoing arrows and Finish three incoming arrows, but every route must pass through one of the two arrows leaving D. Hence no collection of three fits. The board has 18 individual routes and 36 unordered maximum collections, but listing them is unnecessary. See the reference diagram on page 7. Hint: look in the middle, not only at the ends.

Problem 3 and K-1 Problem 5

All eight pairs of closing arrows on the bowtie are below. Here brackets mean “choose one from each listed pair of alternatives.” Before C, choose one from {sA,AC} and one from {sB,BC}. After C, choose one from {CD,Dt} and one from {CE,Et}.

Before C	After C
{sA,sB}	{CD,CE}
{sA,BC}	{CD,Et}
{AC,sB}	{Dt,CE}
{AC,BC}	{Dt,Et}

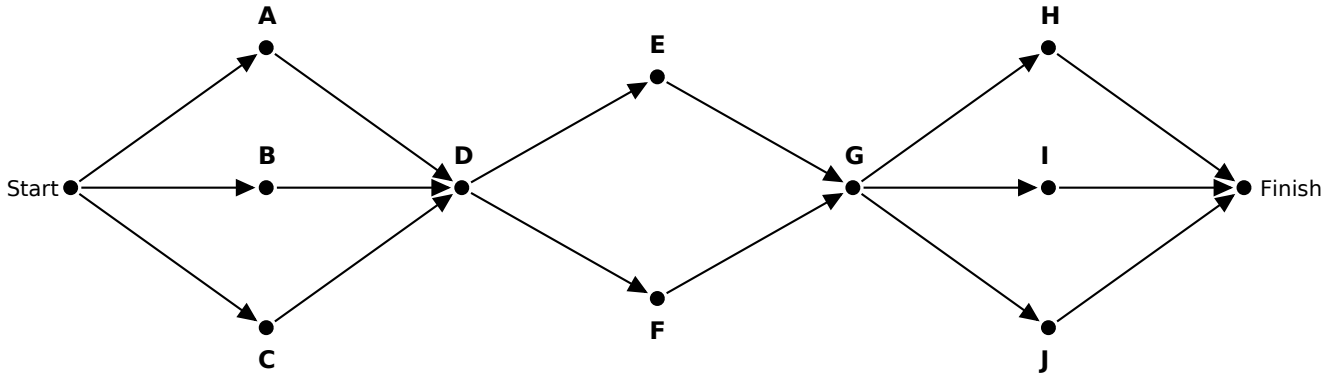
Completeness: if a path from s to C and a path from C to t both survive, concatenate them to get a whole route. So at least one half must be entirely blocked. Each half has two independent branches; blocking it with exactly two arrows requires one closure on each branch. A mixed pair with one arrow on each half cannot block either half. Hint ladder: cover the right half temporarily; ask what it takes to stop reaching C; then repeat on the other half.

Problem 4

Upper board: any new arrow from A, B, C or D to G, H, I or J works (16 choices); it must jump past both two-arrow bottlenecks. With D to G, three routes are s-A-D-E-G-H-t, s-B-D-G-I-t, and s-C-D-F-G-J-t. They meet at D and G but share no arrows. Lower board: impossible. Start has only two outgoing arrows, and the allowed new arrow cannot touch Start. Every collection is therefore bounded by 2. Hint: before searching, ask whether there is a bottleneck the allowed move cannot change.

Middle Problem 5 is fully solved on page 7. Middle Problem 6 is the two-player game solved under K-1 P6 on page 5; the letters name the same arrows.

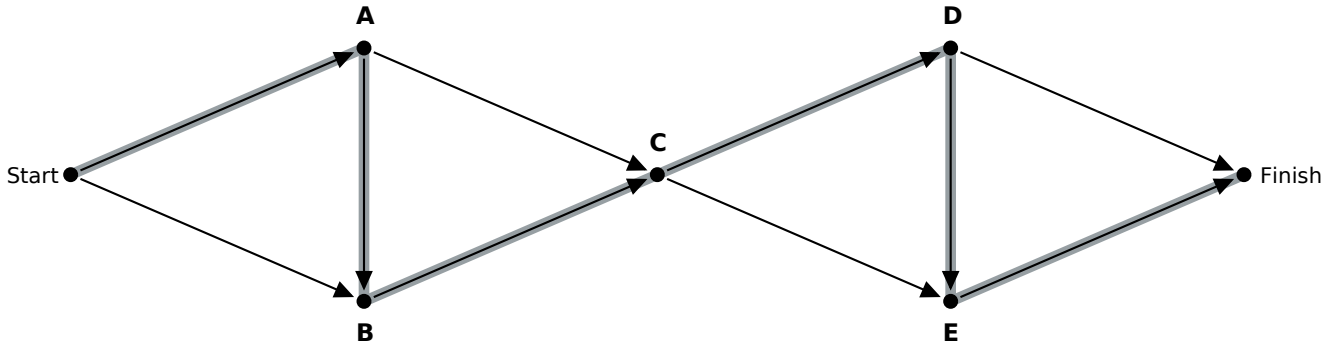
The interior bottleneck and revised rerouting



Reference for middle P2/P4 and upper P2/P6. Two routes and the cut {DE,DF} certify 2. Adding DG makes the middle P4 upper board support 3; removing B and its incident arrows makes the lower P4 board start-limited.

Middle Problem 5 and upper Problem 1

First board: replace the reserved s-A-B-t by s-A-t and s-B-t. The change-walk s-B-A-t uses unused sB and At and cancels AB by traversing it backward. Two Start arrows certify the optimum of 2.



Second board is the revised coupled network, with the reserved route s-A-B-C-D-E-t shown thick above. Its maximum is 2, not 3. Two final collections are {s-A-C-D-t, s-B-C-E-t} and {s-A-C-E-t, s-B-C-D-t}. No third can leave Start.

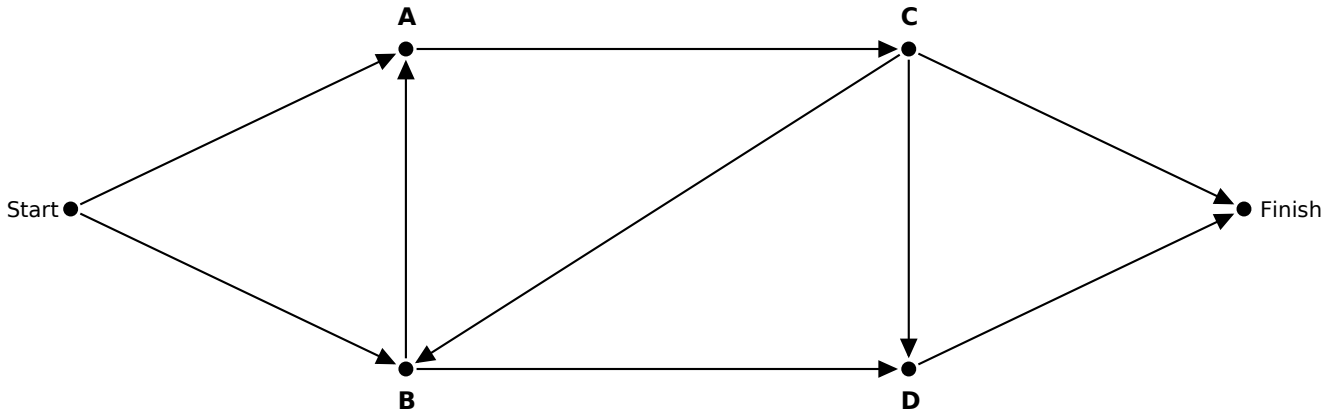
The unique simple change-walk is s-B-A-C-E-D-t. It adds sB, AC, CE, Dt and cancels AB and DE. Together with the retained sA, BC, CD, Et, these form the two bowtie collections above. At C the two incoming units can be paired with the two outgoing units in either way. Both cancellations are necessary in this residual walk; doing only one still leaves the other half blocking an increase.

Held hints: First ask whether both arrows leaving Start could be used. Then ask which reservation prevents the other approach from continuing. If that repair still leaves only one route, inspect the other half. Do not call a backward change-step a legal final route arrow. The final packets include clean working copies on middle page 6 and upper page 2.

Grades 4-5 Problems 2 to 4

Problem 2

Use the interior bottleneck solution on page 7: two routes and two closures, DE and DF. Every s-to-t route must leave the start-side set {s,A,B,C,D}, and these are its only outgoing arrows. The certificate argument on page 4 supplies both required inequalities. Hint: explain why all routes cross the proposed barrier before claiming the count is optimal.



Problem 3

Any two of these three start-side sets work. All other dots are on the Finish side. They are the complete list satisfying both requested conditions.

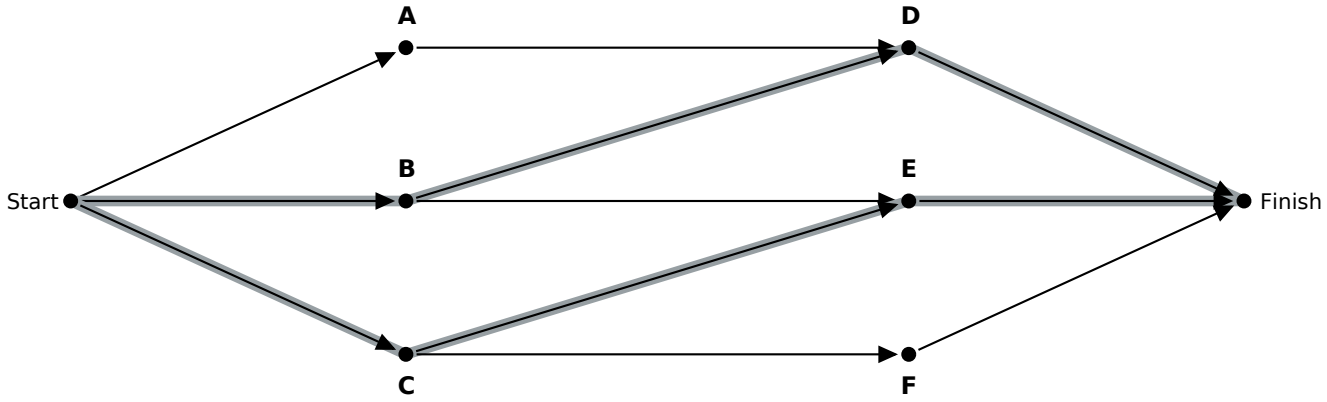
Start-side set	Outgoing cut arrows	Arrows entering Start side
{s,A}	sB, AC	BA
{s,A,B}	AC, BD	CB
{s,A,B,D}	AC, Dt	CD, CB

Every directed route begins on the Start side and ends outside it, so at its first exit it must use an outgoing cut arrow. Closing all outgoing cut arrows therefore blocks travel. Incoming arrows do not allow an exit and are not counted. A route can later reenter the side, but that cannot avoid its first exit. Hint: color the dots by side, then follow each arrowhead across the color boundary.

Problem 4

Close AC and BD. This is a minimum two-arrow set: s-A-C-t and s-B-D-t are disjoint, so one closure is insufficient. The route s-A-C-B-D-t uses both AC and BD. It cannot belong to a largest collection: every other route must use AC or BD, both already reserved. The largest size is 2, but any collection containing this long route has size 1. This example refutes the claim that every route uses exactly one outgoing cut arrow; “at least one” is the correct bound. There are five smallest sets: {sA,sB}, {sB,AC}, {AC,BD}, {AC,Dt}, {Ct,Dt}. Only the middle three are used twice by one route (s-B-A-C-t, s-A-C-B-D-t, s-A-C-D-t). If a child chose the two Start or the two Finish arrows, ask for a different smallest set. The answer to the question is always no. Hint: reserve the long route and test which cut arrows remain available.

Grades 4-5 Problems 5 and 6



Problem 5

The reserved routes are s-B-D-t and s-C-E-t. Use the change-walk s-A-D-B-E-C-F-t. The steps D to B and E to C cancel the original BD and CE. Other steps reserve sA, AD, BE, CF, Ft. The resulting three forward routes are s-A-D-t, s-B-E-t and s-C-F-t. All arrows are present in the printed board. The three arrows leaving Start bound every collection by 3, so this improvement is optimal. The problem also asks for the two Problem 1 boards: their change-walks are s-B-A-t and s-B-A-C-E-D-t (page 7).

The change-walk is forced here: only sA is initially available from Start; after AD, the used Dt is unavailable forward, so go backward along BD; continue analogously through E. This is a different interaction from the revised bowtie case. Hint: redraw only the currently available forward steps and allowed cancellation steps, keeping their meanings distinct.

Problem 6

The printed reservations on the bottleneck board are s-A-D-E-G-H-t and s-C-D-F-G-J-t. Exactly s,A,B,C,D are reachable by change-walk steps. A search witness: s-B-D reaches B and D; backward steps D-A and D-C then reach A and C. No change-step leaves that set because DE and DF are reserved and point outward; their backward directions start outside the set.

Thus {DE,DF} is a two-arrow closure matching the two reserved routes. No other dot is reachable: E,F,G,H,I,J,t all lie beyond those saturated arrows, and there is no unused outgoing crossing. Hint: mark reached dots, examine every allowable step leaving one, and continue until no new dot can be marked. Do not stop after one failed branch.

Common misconceptions to catch

A greedy route can use both scarce exits. “I cannot add a route” tests only unused forward arrows; it does not establish optimality. A reverse change-step cancels an old reservation; final routes never move against a printed arrow. Incoming edges of a cut do not add to the outgoing capacity. Sharing C, D or G is legal. Backward arrows can create a directed cycle, but a student route still cannot revisit a vertex.

Grades 4-5 Problem 7 and the general proof

Answer: yes. Suppose a collection of k routes has no change-walk to Finish. Mark S, all vertices reachable from Start by unused forward arrows and used backward arrows. Start is in S and Finish is outside. Every original arrow leaving S is reserved, or its unused forward direction would reach another dot. Every original arrow entering S is unreserved, or its backward direction would reach a dot outside S.

At every internal dot, the number of reserved arrows coming in equals the number going out. Add this balance over all dots of S. Internal arrows cancel in pairs. The only surplus is k at Start, since Finish is outside. Hence k equals the number of reserved outward arrows minus the number of reserved inward arrows. The latter is zero; all outward arrows are reserved. There are therefore exactly k outward arrows, and closing them blocks every route. Page 4's certificate proves the current collection is largest.

For a child's drawing proof, put a + token at the tail and a − token at the head of each reserved arrow, and total the tokens on the dots of S. Dot by dot: each internal dot has as many + as −, and Start has k +, so the total is k. Arrow by arrow: an arrow inside S gives one + and one −, an arrow leaving S one +, and an arrow entering S would give one −, but none is reserved. So exactly k arrows leave S, all reserved.

Why equal optima exist

Begin with no reservations. If a change-walk reaches Finish, toggle its arrows: add unused forward arrows and cancel used backward arrows. Every internal visit preserves equal inflow and outflow; Start gains one outgoing unit and Finish one incoming unit. The value increases by one, never exceeds the number of arrows leaving Start, and stays an integer. Thus after finitely many improvements there is no such walk and the matching cut argument applies.

Toggling may leave directed cycles. Delete their reserved edges, which preserves balance and the value, then trace remaining units from Start to Finish. If tracing repeats a vertex, remove that cycle first. Finite tracing decomposes the integral reservation into edge-disjoint simple routes. This closes the gap between a flow and the packet's route definition; examples alone do not prove the general theorem.

Extensions and limits

Ask children to invent a graph where a first route gets stuck, trade it with another table, and provide a matching cut for the repaired optimum. Or ask which one-arrow additions change the bottleneck count. Keep one common Start and Finish, directed unit-capacity edges, and edge-disjoint routes. The argument here does not establish a theorem for several prescribed endpoint pairs or vertex-disjoint routes.

Sources and verification

Robert Sedgewick and Kevin Wayne, Algorithms, §6.4 "Maximum Flow," slides 20-26, 28-37 and 40: https://www.cs.princeton.edu/courses/archive/fall15/cos226/lectures/64MaxFlow.pdf. Official companion page, exercise 2 and exercises 5-6: https://algs4.cs.princeton.edu/64maxflow/. These support the flow/cut framework; final board instances, exact keys and proofs were checked for this packet.

Teaching context: Rozhkovskaya, Math Circles for Elementary School Students, "Introduction: Berkeley 2009," reports additional instructors and active parent help. The launch, timing and held-hint choices here are unpiloted local suggestions. Independent exhaustive paths, packings, closures, game states and residual reachability are saved in facilitator-src/math-checks.json. In particular the revised board has 9 routes, 2 maximum collections and one simple augmenting walk requiring both cancellations. Upper P5 adds a cancellation example.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

For another visit, keep one best route collection and its matching closure certificate. K-1 can find every preserving closure (P4) or every stopping pair (P5), or play the closure game (P6). Grades 2-3 can choose complete closures (P3), an added-arrow puzzle (P4), or rerouting (P5). Grades 4-5 can return to cuts and change-walks (P3-6), then use P7 as a separate general-proof investigation. The proof route remains available even if the first meeting ends with one physical certificate.